Fundamentals of Progressive Addition Lens Design

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BACKGROUND

It is hard to believe that the concept of a *progressive addition lens* (*PAL*) has been around since 1907.[1] Although early progressive lenses were rather crude in design, they have consistently improved in both performance and sales over the past few decades. Recent figures from the Optical Industry Association suggest that progressives lens sales are growing rapidly, and now represent approximately 38% of all multifocal lenses sold in the US.[2] Even so, the principles behind progressive addition lenses still remain somewhat of a mystery to many eyecare professionals.

Conventional progressive addition lenses are one-piece lenses that vary gradually in surface curvature from a minimum value in the upper, *distance* portion, to a maximum value in the lower, *near* portion. The result is a smooth, continuous increase in surface power that provides the necessary near addition (or add power), without any visible lines of demarcation or abrupt disturbances of vision. Figure 1 below demonstrates the gradual increase in curvature and surface power towards the lower, near portion of the lens.

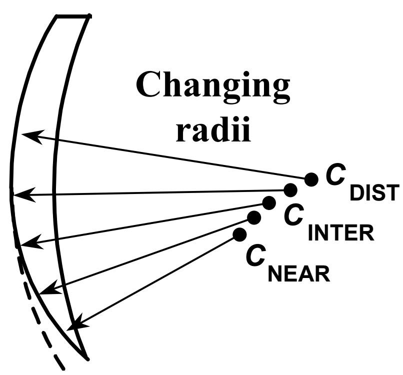

Figure 1. Cross-sectional representation of a progressive lens surface. The shorter radius of curvature in the near portion at point C_{NEAR} provides a stronger surface power than the longer radius of the distance portion at C_{DIST}.

A typical, general-purpose progressive lens will have three distinct zones of vision:

1. **Distance**. A designated zone located in the upper portion of the lens, which provides the necessary distance correction.
2. **Near**. A designated zone in the lower portion of the lens, which provides the required near addition (or add power).
3. **Intermediate**. A 'corridor' in the central portion of the lens connects these two zones, which increases progressively in plus power from the distance to near.

These three zones of vision blend together seamlessly, providing the wearer with a continuous depth of field from near to far, as illustrated in Figure 2.

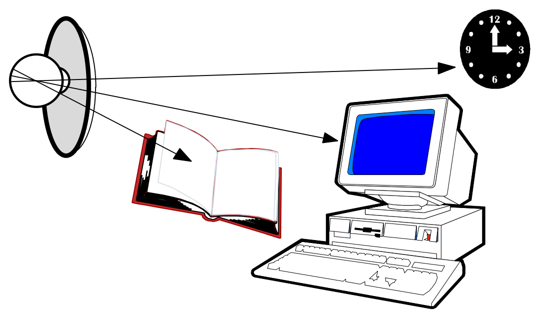

Figure 2. The three ranges of vision through a progressive addition lens.

Several methods exist for evaluating the optics of progressive addition lenses. One particularly convenient method of describing progressive lens surfaces is the **contour map**. Contour maps are similar to topographical maps, and use lines connecting points of equal power on the lens surface. Each contour line/shade represents an increasing level of power at a given interval, usually 0.50 D.

The gradual increase in power provided by a progressive lens surface can be described by either a **mean power plot** or a **power profile plot**. Both quantify the change in power, either by mapping the zones of increasing power, or by plotting it as the power changes along the progressive corridor (or **umbilical line**). The lens surface depicted in Figure 3 below has a +2.00 D add power.

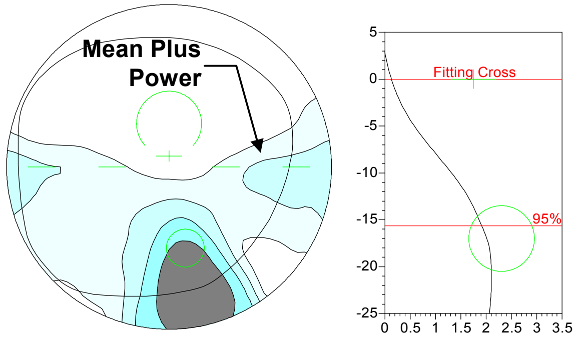

Figure 3. A contour plot is similar to a topographical map. This mean power plot shows the gradual increase in plus power towards the near portion of the lens. Each contour level represents 0.50 D of additional plus power.

Contour plots, although a useful tool for analyzing and comparing the optics of a progressive addition lens, are simply mathematical models of the lens surface. Although they may be indicative of lens performance, they are not enough to predict patient acceptance.

Most eyecare professionals are aware of the product benefits afforded by the optical features of a progressive addition lens:

- No visible segments or lines of demarcation—provides more cosmetically appealing lenses with continuous vision, free from visually distracting borders;
- Clear vision at all distances—provides vision that more closely resembles the lost accommodation of the eyes; and
- No unwanted image displacement—or *jump*—ensures that there are no abrupt disturbances of vision.

Every progressive lens design requires a globally smooth surface that provides a gradual transition in curvature from the distance portion down into the near portion. Further, this gradual blending of curvature means that the add power is gradually changing across a large area of the lens surface, not just at the bottom (see Figure 3).[3]

Unfortunately, the superior optics and line-free nature of a PAL surface *does* have a bit of a price to pay... This change in curvature results in an inevitable consequence: surface astigmatism.

Surface astigmatism produces an unwanted **astigmatic error** (or **cylinder error**) that can, in sufficient quantities, blur vision and limit the wearer's field of clear vision. Therefore, this astigmatic error essentially serves as a boundary for the various zones on the progressive lens surface. Unwanted astigmatism, which is a consequence of the lens design, is influenced by:

- **Add power**. The amount of astigmatism will be directly proportional to the add power of the lens. A +2.00 D add, for example, will generally produce twice as much astigmatic error as a +1.00 D add.
- **Length of the progressive corridor**. Shorter corridors produce more rapid power changes and higher levels of astigmatism, but reduce the eye movement required to reach the near zone. Longer corridors provide more subtle power changes and lower levels of astigmatism, but increase the eye movement required to reach the near zone of the lens.
- **Width of the distance and near zones**. Wider distance and near zones confine the astigmatism to smaller regions of the lens surface, which produces higher levels of this astigmatism. Wider zones do provide larger areas of clear vision, however.

A well-designed progressive lens will reduce the amount of astigmatic error to its mathematical limits for a given design. During the design and optimization process, various parameters are adjusted to control and manipulate the distribution and magnitude of this astigmatic error across the progressive lens surface. The width of the near and distance zones, and the length of the progressive corridor, are the chief parameters that are altered, as shown in Figure 4.

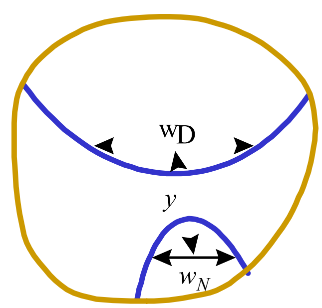

Figure 4. w_D is the width of the distance zone at a specified height; w_N is the width of the near zone at a specific depth; and y is the length of the progressive corridor—or umbilical line—connecting the two.

To understand how the length of the progressive corridor and the add power can affect the rate of change and magnitude of the astigmatic error, consider a progressive lens with a +2.50 D add, and a 17-mm progressive corridor. This lens will have to change by 2.50 D over a distance of 17 mm. This represents an average change of roughly: 2.50 / 17 = 0.15 D per millimeter.

Consequently, the power of a progressive lens changes more rapidly down the progressive corridor as the add power increases, or as the length of the corridor decreases. The magnitude, distribution, and rate of change (or *gradient*) of this astigmatism are all performance factors that can affect the wearer's acceptance of the lens.

Progressives are often arbitrarily classified into two large categories, or design 'philosophies,' by the relative magnitude and distribution of their surface astigmatism:[4]

- **Harder Designs**. A *harder* PAL design concentrates the astigmatic error into smaller areas of the lens surface, thereby expanding the areas of perfectly clear vision at the expense of higher levels of blur and distortion. Consequently, harder PALs generally exhibit four characteristics when compared to softer designs: wider distance zones; wider near zones; shorter, more narrow progressive corridors; and higher, more rapidly increasing levels of astigmatic error. See Figure 5.

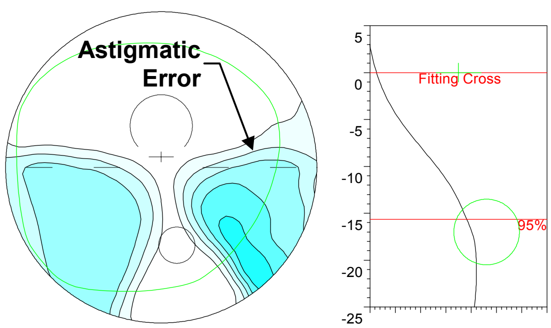

Figure 5. Harder design.

- **Softer Designs**. A *softer* PAL design spreads the astigmatic error across larger areas of the lens surface, thereby reducing the overall magnitude of blur at the expense of narrowing the zones of perfectly clear vision. The astigmatic error may even encroach well into the distance zone. Consequently, softer PALs generally exhibit four characteristics when compared to harder designs: narrower distance zones; narrower near zones; longer, wider progressive corridors; and lower, more slowly increasing levels of astigmatic error. See Figure 6.

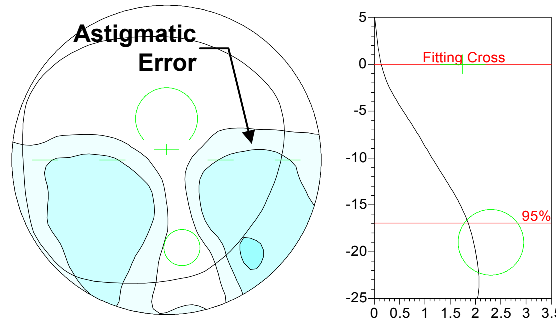

Figure 6. Softer design.

In general, harder PAL designs will provide wider fields of view, and will require less head and eye movement, at the expense of more swim and blur. Softer PAL designs will provide reduced levels of astigmatism and swim, while limiting the size of the zones of clear vision and requiring more head and eye movement. Modern PALs are seldom absolutely 'hard' or absolutely 'soft.' Unfortunately, such terms do not satisfactorily describe modern lenses. Many of the recent PAL designs incorporate the best balance between these two design philosophies. In the future, more accurate language will most likely be developed.

Early progressive lenses were *symmetrically* designed so that the right and left lenses were identical. To achieve the desired inset for the near zone, the lens blanks were simply rotated 9 or 10°. The principal drawback to this was the disruption of binocular vision as the wearer gazed laterally across the lens, since the astigmatism differed between the nasal and temporal sides of the distance zone. Many modern lens designs, however, are *asymmetrically* designed with separate right and left lens designs. The amount of astigmatic error on either side of the progressive corridor can now be adjusted independently. Compare the distance zones in the early *symmetrical* right and left designs in Figure 7 with the *asymmetrical* designs in Figure 8.[5]

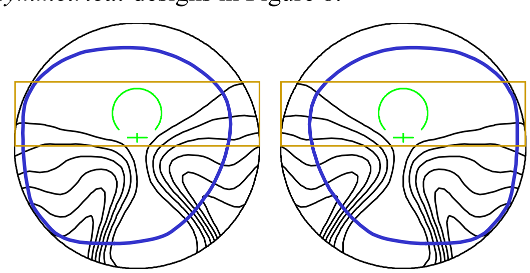

Figure 7. Early *symmetrical* PAL design in which the distortion rises into the *nasal* distance zone on both lenses after they have been rotated.

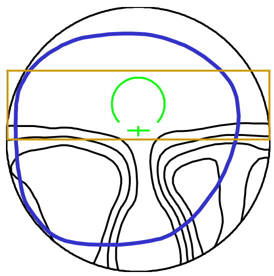

Figure 8. Newer *asymmetrical* PAL design.

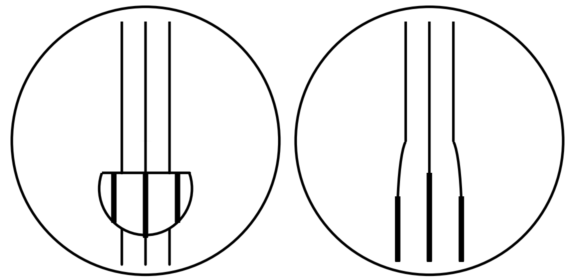

Figure 9. The change in power and magnification through the progressive corridor and near zone produces skew distortion, when compared to conventional multifocals.

About ten years ago, some manufacturers began to vary the lens design from one add power to another. This was a **multi-design** concept that employed add-specific lens designs to customize the PAL designs for the various stages of presbyopia. For example, the design employed for a +2.50 add power varies slightly from the design employed for a +1.50 add.[6]

Recently, SOLA Optical introduced another concept: **Design by Prescription™**; the use of different PAL lens designs for different distance refractive errors. This philosophy is currently employed in the Percepta® progressive addition lens. For example, the design employed for the 7.25 base curve varies slightly from the design employed for the 5.25 base.

In addition to the astigmatic error inherent in a progressive lens, the change in power and magnification produced by the corridor and near zone of the lens results in **skew distortion**. Objects, like straight lines, may appear curved—or skewed—when viewed through the lateral areas in the lower portion of the lens. The *more curved* a vertical line appears, the *less* **orthoscopic** the lens is through that particular zone. This effect is illustrated in Figure 9.[7]

Progressive addition lenses are supplied with two types of marking for layout, power verification, dispensing, and identification purposes. *Removable* markings, which are inked on, identify the layout, verification, and dispensing points of the lens. And *permanent* markings, which are engraved upon the surface, provide the identification and add power of the lens, as well as locator marks to reapply the ink markings when necessary. The standardized locations of these markings are shown in Figures 10 and 11. Table 1 provides some additional information about each marking.[8]

We will conclude this article with a few tips on fitting progressives. Proper fitting is crucial to ensure success with PALs, since the zones of clear vision are narrower than conventional multifocals produce. For most general-purpose progressives, the fitting point/cross should be placed directly in front of the pupil, as illustrated Figure 12. Monocular interpupillary distance measurements should always be utilized—preferably with the aid of a corneal reflex pupillometer.

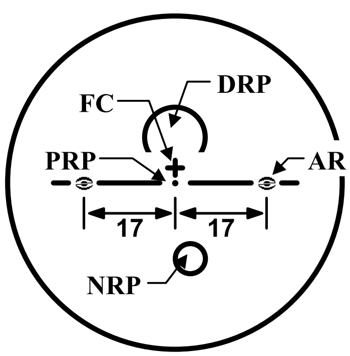

Figure 10. Removable ink markings.

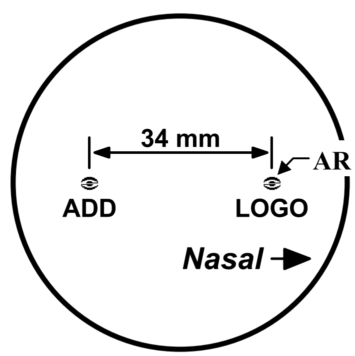

Figure 11. Permanent semi-visible engravings.

Table 1 ISO/ANSI PAL markings

Reference Mark/Location	Abbr.	Type	Purpose
Prism reference point	PRP	Removable	Surfacing layout Prism verification
Distance reference point	DRP	Removable	Distance power verification
Fitting cross (or point)	FC	Removable	Fitting reference Finishing layout
Near reference point	NRP	Removable	Add power verification
Alignment reference marking	ARM	Permanent	Axis alignment Re-marking lenses Identification of lens type or manufacturer
Add power	ADD	Permanent	Identification of add power
Logo or trademark	LOGO	Permanent	Identification of lens type or manufacturer

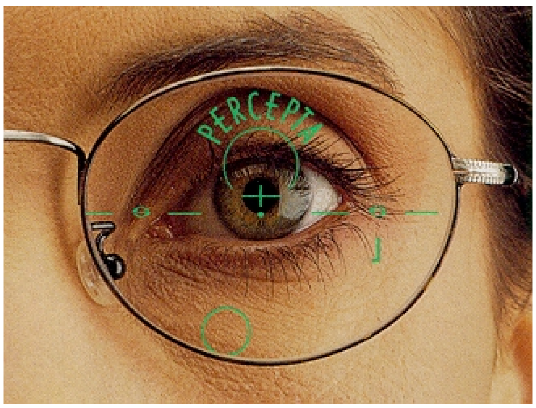

Figure 12. The fitting point/cross should be placed directly in front of the pupil center.

For optimum performance, also remember the frame fitting guidelines below:

PAL Frame Fitting Guidelines

- Remember to pre-adjust the frame *before* taking any measurements.
- Maintain a short vertex distance to maximize the field of view through the various zones of the lenses. 11 to 14 mm is an acceptable range.
- Pantoscopic tilt is also important, and will help maximize the fields of view through the near and intermediate zones. 8° to 14° is an acceptable range.
- Some face form wrap will also be beneficial when fitting progressives.
- The fit and vertical depth of the frame should allow for the manufacturer's minimum recommended fitting height, which is often between 22 to 24 mm.

Educating your patients about the nature of their new progressive lenses will also improve the likelihood of success. If your patients are fully aware of what to expect beforehand, they will be much more apt to tolerate any initial visual discomfort produced by the lenses during the first few days of wear.

References

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